

LexFusion

AUTONOMOUS MULTI-AGENT LEGAL RAG CHAMBER

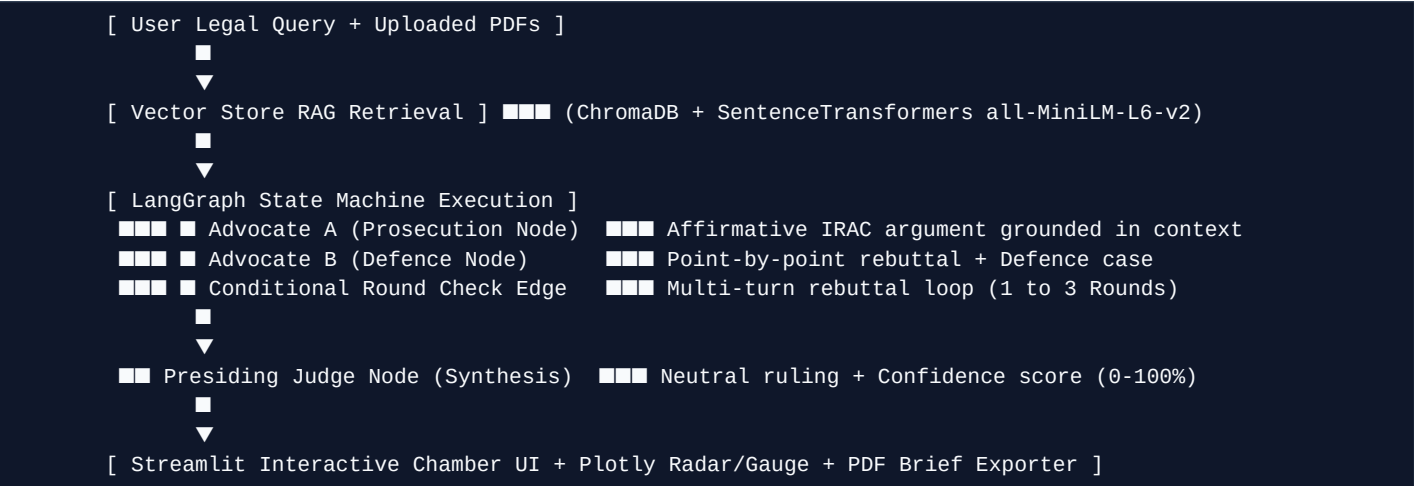
Repository:
github.com/anagha16543/fusion

Branch: main | Version: v2.4.0

Deployment: Streamlit Cloud

Executive Summary:
LexFusion transforms static legal document analysis into an autonomous, multi-agent courtroom simulation. Orchestrated using a **LangGraph StateGraph**, three specialized AI agents (Prosecution, Defence, and Judge) cross-examine legal context, challenge arguments in multi-turn rebuttal loops, eliminate single-shot AI bias, and generate grounded legal rulings complete with quantitative confidence scoring.

1. System Architecture & Workflow



2. Multi-Agent Cross-Examination Engine

Agent	Role & Perspective	Core Function & Methodology
■ Agent 1: Advocate A	Prosecution / Plaintiff	Constructs affirmative legal arguments using the IRAC framework (Issue, Rule, Application, Conclusion) strictly grounded in retrieved evidence. Pre-empts counterarguments.
■ Agent 2: Advocate B	Defence / Respondent	Executes point-by-point rebuttals against Prosecution claims. Identifies evidentiary gaps, procedural issues, or statutory exceptions from retrieved context.
■ Agent 3: Presiding Judge	Impartial Judicial Arbiter	Synthesizes both positions, evaluates evidence quality, parses a quantitative Confidence Score (0-100%) , highlights strengths/weaknesses, and issues a final binding legal opinion.

3. Technology Stack Breakdown

Category	Technologies	Implementation Details
Agent Framework	LangGraph 0.4.x, LangChain 0.2.x	StateGraph execution, state immutability, cycle loop routing.
LLM Inference	Groq API (Llama-3.3-70B), Gemini	Ultra-fast model inference for multi-agent multi-round debates.
Vector Store & Embed	ChromaDB, SentenceTransformers	Semantic PDF chunking (`all-MiniLM-L6-v2`) & similarity scoring.

Backend REST API	FastAPI, Uvicorn, Pydantic v2	Endpoints `/query`, `/upload`, `/health`, `/stats` with strict typing.
Frontend UI	Streamlit 1.35+, Plotly 5.20+, CSS3	Glassmorphic UI (`#050814`), split-screen debate view, sound FX.
Multilingual Engine	Multilingual System Directives	Supports 50 global languages across all 3 AI agent outputs.
Analytics & Export	Plotly Gauges/Radar, Brief Exporter	Visual legal strength metrics, match bars, and HTML/PDF brief generator.

4. Standout Key Innovations

- **Dual-Mode Execution:** Automatic fallback to in-process local agents when FastAPI server is offline, ensuring 100% uptime on Streamlit Cloud.
- **Adversarial Legal Testing:** Eliminates single-agent bias by forcing opposing perspectives to battle over retrieved facts.
- **Zero-Hallucination Grounding:** Agents are constrained to cited document chunks with similarity percentage match bars ([██████████] 94%).
- **Interactive Legal Analytics:** Plotly Legal Strength Radar Chart & Confidence Radial Meter provide instant executive visibility.
- **50 Global Languages:** Full multilingual prompt engineering enables cross-border legal contract cross-examination.